

SLM Mammamety Loyuka Xisob - Kibab

Model: SLM

Architecture: Decoder only transformer

Vocabulary: 50,000 tokens

d-model: 256

Layers: 6

Attention Heads: 8

Sequence Length: 128

Batch Size: 32

FFN dimension: 1024

Epochs: 3

Dataset: 1,000,000 tokens

precision: FP16

Optimizer: AdamW

$$d_{ff} = 4 * 256 = 1024$$

Input Shape

Batch size = 32

Sequence length = 128

Input shape: (32, 128)

In degani 32 ta sample har bir

sample 128 ta token

Butta batchdagi tokenlar soni

$$\text{Token per Batch} = B \times T$$

$$32 * 128 = 4096$$

tokens
+ batch = 4096 tokenlar iborat

Embedding hisoblash

vocab: 10,000

d = 256

Embedding matrix = $V \times d$

Demak $10000 \times 256 = 2,560,000$ parameter

Attention Head shchanch

d_model = 256

heads = 8

har bor head dimension

d_head = d_model / heads

d_head = $256 / 8 = 32$

Demak $8 \times 32 = 256$ head manjud

Q, K, V hisoblash

har bor $256 \times 256 = 65,536$ parameter

agar bor ham shchanch

$3 \times 256 = 768$

shu sababli QKV parameters

= $196,608 + 768 = 197,376$

FF Network

$263,168 + 262,400 = 525,568$

67a transformer block

$188,460 \times 6 = 1,130,676$

Yani: 1,288,072 million parameter